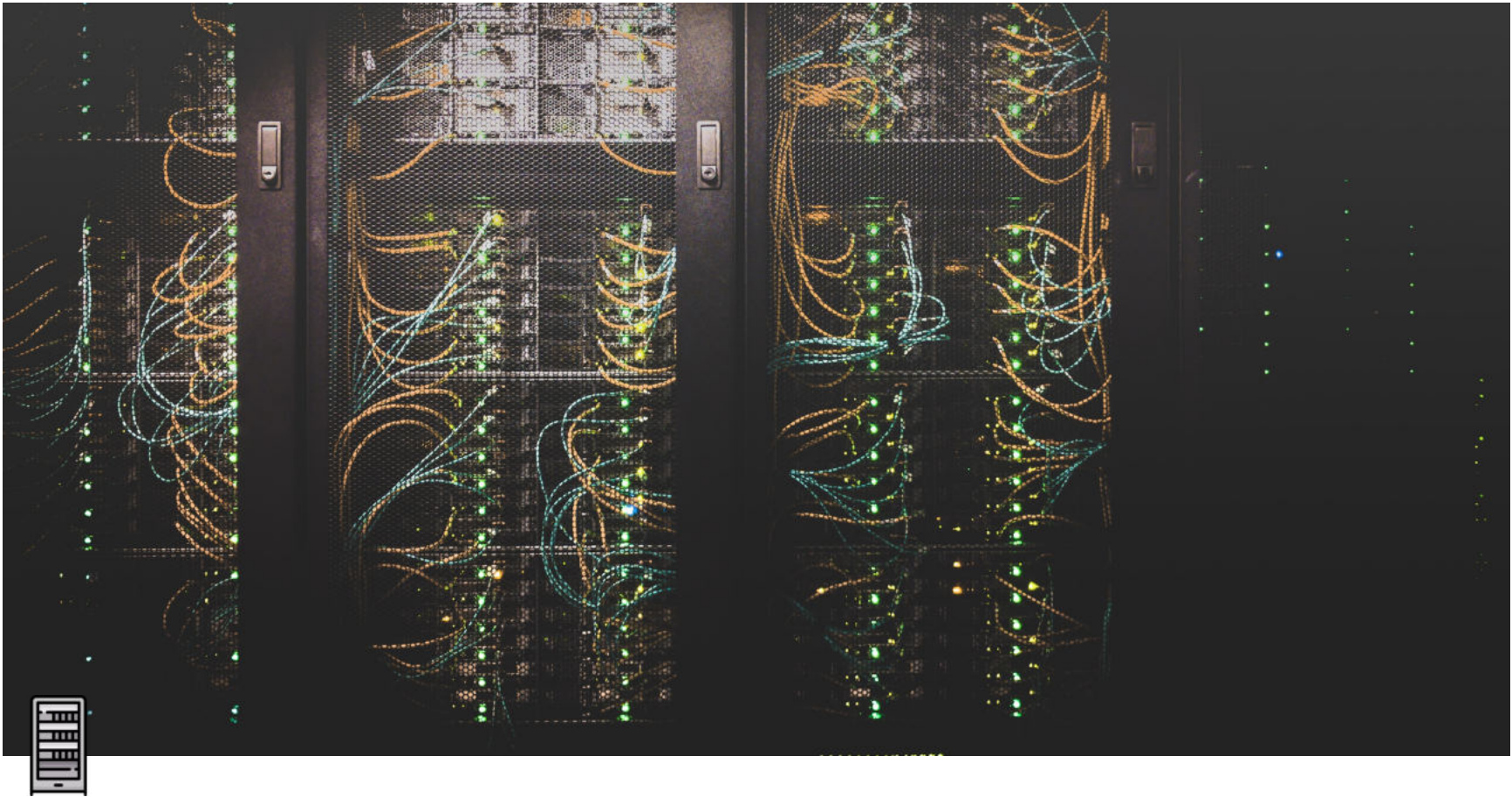




Simple Commands in Linux - 1

Type	Lecture
Date	@December 22, 2021
Lecture #	3
Lecture URL	https://youtu.be/DlpBEmRDHnw
Notion URL	https://21f1003586.notion.site/Simple-Commands-in-Linux-1-3e5b2c25e6d04a7499afcb89af041b20
Week #	1

Some basic commands

- `date` → Date and time

```
kashif@Zen:~/Desktop$ date
Wednesday 22 December 2021 12:05:01 PM IST
```

- `date -R` → Gives the date in RFC5322 standard
- `cal` → Calendar of a month

```
kashif@Zen:~/Desktop$ cal
December 2021
Su Mo Tu We Th Fr Sa
      1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

- `free` → Memory statistics

```
kashif@Zen:~/Desktop$ free
              total        used        free      shared  buff/cache   available
Mem:      4020444       589808       2725032        36244        705604       3170416
Swap:      945368           0         945368
```

- `free -h` → Makes the output human readable
- `groups` → Groups to which the user belongs

```
kashif@Zen:~/Desktop$ groups
kashif adm cdrom sudo dip plugdev lpadmin lxd sambashare
```

idk what the junk is this

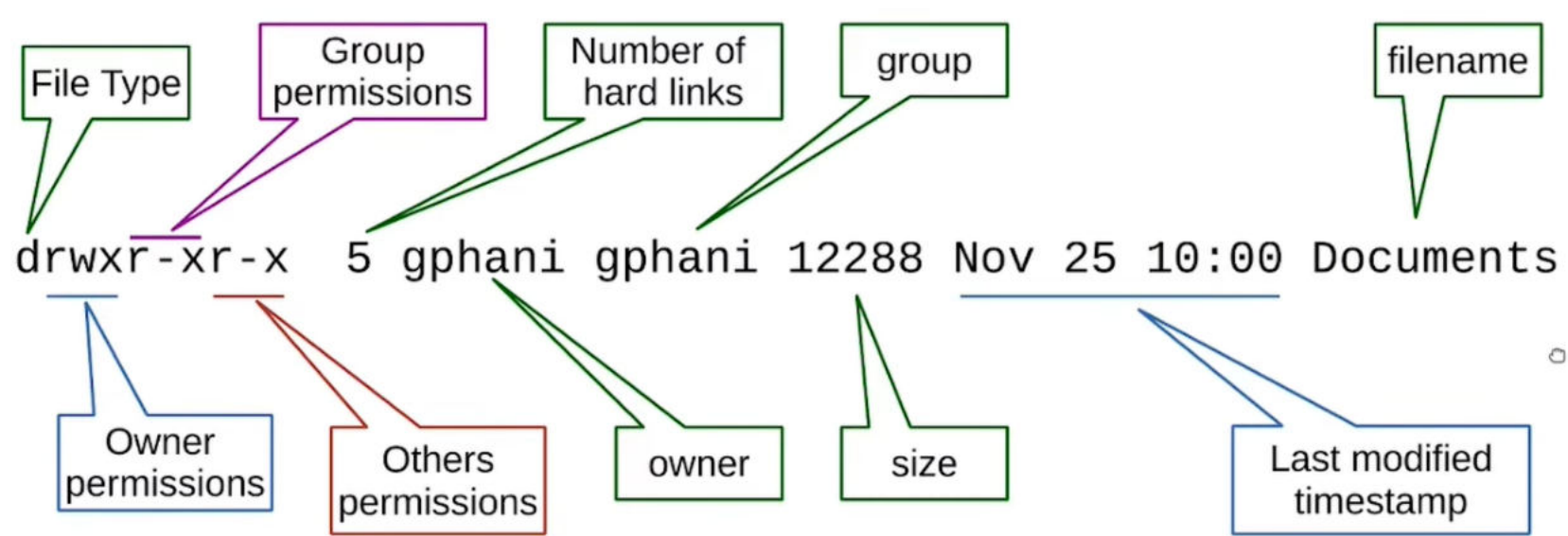
- `file` → What type of a file it is

```
kashif@Zen:~$ file .bashrc
.bashrc: ASCII text
```

- `cd -` → To visit the previous directory we were in

```
kashif@Zen:~$ cd -
/home/kashif/Desktop
kashif@Zen:~/Desktop$
```

Typical output of `ls -l`



File types

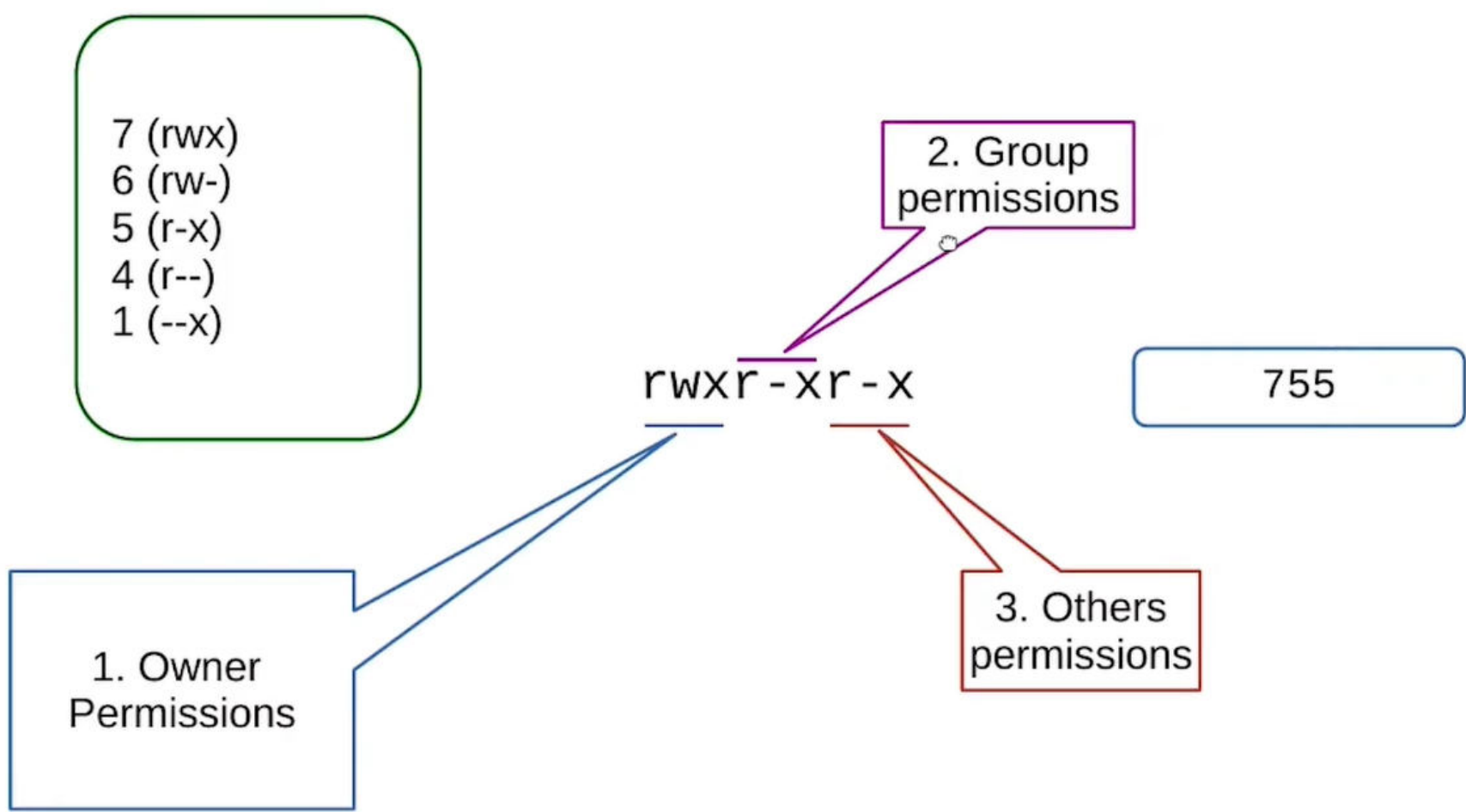
- `-` → Regular file
- `d` → Directory
- `l` → Symbolic link
- `c` → Character file
- `b` → Block file
- `s` → Socket file
- `p` → Named pipe

inode (Read: eye node)

`ls -l <name>`

- An entry in the filesystem table about the location in storage media

Permission string



To modify permissions

- Create a folder
 - `mkdir <folder-name>`
 - `chmod g-w <folder-name>` to remove the write permission from the group
 - Similarly, `chmod g-x <folder-name>` to remove the execute permission from the group
 - To add permission, `chmod g+w <folder-name>` to give write permission to the group
- So, a general structure of permission syntax is something like ...
 - `chmod <user-group><plus/minus><r/w/x> <folder-name/file-name>`
 - Where `<user-group>` are ...
 - `u` → User
 - `g` → Group
 - `o` → Others
 - `<plus/minus>` are ...
 - `-` → To remove permission
 - `+` → To add permission
 - `<r/w/x>` are ...
 - `r` → Read
 - `w` → Write
 - `x` → Execute
- We can also use numerical values for permissions
 - `chmod 700 <folder-name>` to give the `rwx` permission to user only

`touch` command

- Used to modify the timestamp of a file or folder
 - If a file does not exist, it will be created
- `touch <file-name>` to create a new file
 - `chmod 700 <file-name>` to give `rwx` permission to user only

`cp` command

- `cp <file-name> <new-name>` to copy a file to a new name
 - `cp <file-name> <new-path>` can be used to copy a file to a new path

`mv` command

- `mv <file-name> <new-path>` to move a file to a new path
 - `mv <file-name> <new-file-name>` can be used to rename a file

Also, use quotation marks if the file name includes a space

`rm` command

- `rm <file-name>` to remove a file
 - **IT WILL NOT ASK FOR YOUR CONFIRMATION**
 - Just straight up delete



- This is the default behaviour
 - We can pass `-i` flag for the confirm remove prompt

Alias

- We can also set an alias for long commands, for example ...
 - `alias ll="ls -altrhF"`

Know current user

```
whoami
```

Read a text file, page-by-page

```
less <file-name>
```

To know the type of a file

```
file <file-name>
```

Some commands

- `chmod` → Change permissions of a file
- `touch` → Change modified timestamp of a file
- `cp` → Create a copy of a file
- `mv` → Rename/Move a file
- `mkdir` → Create a directory
- `rm` → Remove a file